Pharmacon Writer Calibration Protocol

India pilot: writer-adaptive recognition for English and Roman-script prescription lines, followed by structured extraction, local inventory matching, and patient records

THE PROJECT IS APPROVED IN THIS FORM

The technical contribution is not a generic prescription scanner or a dashboard. It is a system that learns a clinician's connected handwriting from a small, controlled calibration set, proves the improvement on unseen prescription lines, and then converts confirmed results into clinic inventory and patient workflows.

1. The precise research question

Can sixteen writer-specific calibration lines, followed by four actively selected refinement lines, reduce character and field-recognition error on new English or Roman-script prescriptions from the same Indian clinician without memorising a fixed list of medicines?

This question is small enough to test before the planning presentation and substantial enough to anchor the full 17-week project. Isolated A-Z writing is useful only as a diagnostic. The trainable evidence must be line-level writing because joining, spacing, abbreviations, digits, dosage units, and word context are the real problem.

2. What the demonstration must contain

Stage	Required implementation	What it proves
Fixed-form capture	Photograph or scan a known form; locate the writing boxes and crop each line consistently.	The OCR comparison is not distorted by changing layouts.
Generic recognition	Run the same crops through microsoft/trocr-small-handwritten without writer training.	A reproducible baseline.
Writer adaptation	Adapt a small part of the visual encoder using only that writer's 16 enrollment lines. Freeze the decoder.	Improvement comes from the writer's visual style, not a vocabulary shortcut.
Active refinement	Use alignment errors to choose four additional sentences that cover the writer's weak letters and confusion pairs.	The calibration is responsive instead of merely collecting more data.
Structured prescription	Extract medicine, strength, form, dose, frequency, route, duration, and instructions with rules/grammar.	The output is usable by software, not just OCR text.
Human-safe action	Show top-three formulary matches and confidence; require confirmation before inventory or patient records change.	The system fails safely in a medical setting.

SAFETY BOUNDARY

This is decision support, not autonomous dispensing. A low-confidence word, strength, or dosage must be highlighted and withheld from downstream actions until a clinician or pharmacist confirms it. Never use an LLM to silently rewrite an uncertain medicine name.

Collection protocol and experimental split

Use a pseudonymous Writer ID. Do not collect a doctor's name, signature, council registration number, patient details, or real prescriptions at this stage.

1. Minimum collection

Split	Lines per writer	Allowed use	Never use it for
Enrollment	16 fixed lines	Writer-specific adaptation and confusion analysis	Reporting final performance
Active refinement	4 selected lines	A second adaptation step after error analysis	Choosing the final test cases
Validation	4 fixed lines	Early stopping and confidence thresholds	Training or prompt selection
Held-out test	8 fixed lines	Final generic vs adapted comparison	Any training, correction, or threshold tuning

For the first presentation, collect from at least three writers so the pipeline can be demonstrated. The final project should target 8-12 consenting clinical writers if access permits. Report every writer separately as well as the macro-average; do not mix lines from the same writer across the generic training corpus and the writer-specific test set.

2. Capture rules

- Print at 100% scale on plain A4 paper. Do not shrink or fit to page.
- The printed prompt remains above the blank box and outside the OCR crop. Store the prompt as the ground-truth transcription.
- Ask the writer to write naturally, at their usual speed, with their usual pen. Do not ask for careful block capitals.
- Capture the full page and the cropped line. Keep a ruler or page corners visible in the full image for geometric correction.
- Use one normal-lit image per page for the controlled comparison. Add angle and low-light variants only as a separate robustness test.
- Preserve punctuation, decimal points, slash characters, hyphens, units, and casing exactly in the ground truth.
- Run enrollment and held-out test in separate sessions when possible. This makes the claim more realistic.

3. Required manifest

Field	Example	Purpose
writer_id	W003	Pseudonymous identity
session_id	S1	Separates enrollment from later testing
line_id / split	E07 / enrollment	Prevents leakage
transcription	Tab Amlodipine 5 mg...	Ground truth
page_image / crop_image	W003_S1_P1.jpg / E07.png	Reproducibility
device / lighting / angle	Pixel 7 / indoor / 4 deg	Capture audit
review_status	double-checked	Ground-truth quality

IMPORTANT

Assign the split before scanning. The test lines remain sealed until the pipeline and confidence threshold have been fixed.

Enrollment sheet 1 of 4

Writer ID: _____ Session ID: _____ Date: _____ Pen type: _____

INSTRUCTION TO WRITER

Copy each complete line once, in your normal prescription handwriting and at your usual speed. Do not write in block capitals unless that is your normal style.

E01. The quick brown fox jumps over the lazy dog. 0123456789.

Write the sentence once in your normal prescription handwriting

E02. Sphinx of black quartz, judge my vow. Rx No. 24/09.

Write the sentence once in your normal prescription handwriting

E03. Tab Paracetamol 500 mg, 1 tablet after food, BD for 3 days.

Write the sentence once in your normal prescription handwriting

E04. Cap Amoxicillin 500 mg, 1 capsule TDS for 5 days.

Write the sentence once in your normal prescription handwriting

Enrollment sheet 2 of 4

Writer ID: _____ Session ID: _____ Date: _____ Pen type: _____

INSTRUCTION TO WRITER

Copy each complete line once, in your normal prescription handwriting and at your usual speed. Do not write in block capitals unless that is your normal style.

E05. Tab Azithromycin 250 mg, 2 tablets today, then 1 OD for 4 days.

Write the sentence once in your normal prescription handwriting

E06. Tab Metformin 500 mg, 1-0-1 with meals for 30 days.

Write the sentence once in your normal prescription handwriting

E07. Tab Amlodipine 5 mg, 1-0-0 every morning.

Write the sentence once in your normal prescription handwriting

E08. Tab Levothyroxine 50 mcg, once daily before breakfast.

Write the sentence once in your normal prescription handwriting

Enrollment sheet 3 of 4

Writer ID: _____ Session ID: _____ Date: _____ Pen type: _____

INSTRUCTION TO WRITER

Copy each complete line once, in your normal prescription handwriting and at your usual speed. Do not write in block capitals unless that is your normal style.

E09. Syp Cetirizine 5 mg/5 mL, 10 mL HS for 5 nights.

Write the sentence once in your normal prescription handwriting

E10. Eye drops Moxifloxacin 0.5%, 1 drop QID for 5 days.

Write the sentence once in your normal prescription handwriting

E11. Inhaler Budesonide 200 mcg, 2 puffs BD; rinse after use.

Write the sentence once in your normal prescription handwriting

E12. Ointment Mupirocin 2%, apply locally TDS for 7 days.

Write the sentence once in your normal prescription handwriting

Enrollment sheet 4 of 4

Writer ID: _____ Session ID: _____ Date: _____ Pen type: _____

INSTRUCTION TO WRITER

Copy each complete line once, in your normal prescription handwriting and at your usual speed. Do not write in block capitals unless that is your normal style.

E13. Tab Ibuprofen 400 mg SOS after food; maximum 3 doses/day.

Write the sentence once in your normal prescription handwriting

E14. Cap Cholecalciferol 60,000 IU, once weekly for 8 weeks.

Write the sentence once in your normal prescription handwriting

E15. Tab Telmisartan 40 mg 1-0-0; Atorvastatin 10 mg 0-0-1.

Write the sentence once in your normal prescription handwriting

E16. BP 128/76 mmHg; pulse 82/min; SpO2 98%; temperature 99.4 F.

Write the sentence once in your normal prescription handwriting

Validation and active-refinement sheet

Writer ID: _____ Session ID: _____ Date: _____

Validation lines

These four fixed lines are not used for adaptation. Use them only for early stopping and the confidence/abstention threshold.

V01. Tab Pantoprazole 40 mg, 1 tablet AC breakfast for 14 days.

Write here

V02. Tab Doxycycline 100 mg, 1-0-1 after food for 7 days.

Write here

V03. Inj Insulin 10 IU SC before breakfast and 6 IU before dinner.

Write here

V04. ORS 200 mL after each loose stool; review if symptoms persist.

Write here

Active-refinement lines

These are fallbacks for the first demonstration. In the actual system, generate four lines after inspecting the enrollment errors so they cover the writer's weak characters, frequent confusion pairs, and uncovered prescription tokens.

R01. Mix 0.5 mL with 2.5 mL saline; use once at 8:00 PM.

Write here

R02. Tab Ofloxacin 200 mg, one tablet BD for 5 days.

Write here

R03. Cap Omeprazole 20 mg, one capsule before breakfast.

Write here

R04. Review in 2 weeks; continue current medicines unless advised.

Write here

Held-out test sheet 1 of 2

Writer ID: _____ Session ID: _____ Date: _____

KEEP SEALED

Do not show, scan, or use these lines until adaptation choices and confidence thresholds have been fixed. Test pages must never be added to training or used to choose prompts.

T01. Tab Paracetamol 650 mg SOS after food; maximum 3 tablets/day.

Write the sentence once in your normal prescription handwriting

T02. Syp Amoxicillin 250 mg/5 mL, 5 mL TDS for 5 days.

Write the sentence once in your normal prescription handwriting

T03. Tab Metformin SR 500 mg, one tablet with the evening meal.

Write the sentence once in your normal prescription handwriting

T04. Tab Cetirizine 10 mg, one tablet HS for 7 nights.

Write the sentence once in your normal prescription handwriting

Held-out test sheet 2 of 2

Writer ID: _____ Session ID: _____ Date: _____

KEEP SEALED

Do not show, scan, or use these lines until adaptation choices and confidence thresholds have been fixed. Test pages must never be added to training or used to choose prompts.

T05. Tab Quetiapine 25 mg, 0-0-1 at bedtime for 14 days.

Write the sentence once in your normal prescription handwriting

T06. Tab Clopidogrel 75 mg, one tablet OD after lunch.

Write the sentence once in your normal prescription handwriting

T07. Cap Pregabalin 75 mg, one capsule HS for 10 days.

Write the sentence once in your normal prescription handwriting

T08. Eye drops Tobramycin 0.3%, 1 drop TDS for 5 days.

Write the sentence once in your normal prescription handwriting

India pilot: scope, terminology, and model shortlist

The first version must solve one defined problem well. It should not claim support for every Indian language or every prescription layout.

1. First-release scope

• Recognise English and Roman-script handwritten prescription lines used in Indian clinics. Treat Devanagari, Punjabi, Bengali, Tamil, Telugu, and other scripts as separate later experiments that require their own labelled data and recognition branch.

• Include common dosage forms and shorthand: Tab, Cap, Syp, Inj, drops, ointment, inhaler, OD, BD, TDS, QID, HS, SOS, AC, PC, SC, IM, IV, mg, mcg, g, mL, IU, percentages, decimal strengths, and 1-0-1 schedules.

• Use generic medicine names in the fixed calibration prompts. Add brand aliases only from the participating clinic or pharmacy's own inventory export.

• Keep diagnosis prediction outside the first release. The OCR system must not infer a medicine that the clinician did not write.

2. Indian medicine vocabulary

Build a versioned local formulary in this order: the partner clinic or pharmacy inventory; the National List of Essential Medicines 2022; the PMBJP Product and MRP list; and the National Formulary of India 2021. Store generic name, strength, dosage form, pack size, local brand alias, and inventory SKU. Do not use live web search to decide a medicine at inference time.

3. Models to test

Model	Role in this project	Reason
microsoft/trocr-small-handwritten (62M)	Preferred writer-adaptation model	Line-level HTR, manageable for GPU fine-tuning, and suitable for a small per-writer adapter.
microsoft/trocr-base-handwritten (334M)	Accuracy reference only	Useful to test whether more capacity helps, but too heavy to make the primary deployable model.
PP-OCRv5_mobile_rec (16 MB)	Deployment baseline	Small local model with explicit complex English-handwriting support. Fine-tune on the same line crops.
PP-OCRv5_server_rec (81 MB)	Stronger PaddleOCR comparator	A practical clinic-server baseline with official custom-training support.
CRNN + BiLSTM + CTC	Student-designed baseline	Train through PaddleOCR or MMOCR. It is transparent and deployable, though it may trail transformer models.

RECOMMENDED PATH

Use fixed-form line crops. Run all five generic baselines on the sealed samples. Keep TrOCR-small as the main writer-adaptation branch and PP-OCRv5-mobile as the compact deployment branch. The project contribution is the calibration, adaptation, safe formulary resolution, and measured improvement, not calling an OCR API.

DO NOT USE AS THE FINAL SYSTEM

Tesseract may be shown as a weak baseline for printed text. A hosted OCR API, ChatGPT, Gemini, Qwen-VL, or another vision-language model may be shown only as a comparison. None of these should be the core system, and no model may silently repair an uncertain medicine or dosage.

AI proof-of-concept: exact implementation

Build the smallest experiment that can prove or disprove writer adaptation. Do not begin with the complete clinic dashboard.

1. Baseline

- Rectify the photographed sheet using its page corners, then crop each known writing box. Resize while preserving aspect ratio.
- Use microsoft/trocr-small-handwritten as the unchanged baseline. Save its raw text and token confidence for every line.
- Calculate character error rate (CER) and word error rate (WER) from the stored ground truth. Keep punctuation and units in scoring.

2. Writer-specific adaptation

- Create one adapter per writer. Train only on E01-E16. A practical first choice is LoRA rank 8 on the query and value projections of the last two visual-encoder attention blocks. If that is difficult, fine-tune only the final visual-encoder block.
- Freeze the text decoder in the first experiment. This prevents the model from appearing to improve merely because it memorised the calibration medicine names.
- Use V01-V04 for early stopping. Start with 20-40 epochs, learning rate 1e-4 for LoRA, batch size 4, and strong image augmentation only on enrollment crops: small rotation, perspective, brightness, blur, and scale.
- Save a writer-independent base model and a very small writer adapter. Loading an adapter at sign-in is part of the product design.

3. Build the writer confusion profile

- Align each baseline prediction with its ground truth using Levenshtein alignment.
- Count substitutions as truth character to predicted character, plus insertions and deletions. Normalise each truth-character row to obtain a confusion probability.
- Retain only confusions seen at least twice in the small proof-of-concept; otherwise label them provisional.

4. Select four refinement prompts

Score each unused candidate sentence using the following weighted coverage. Select greedily, reducing the score of characters and tokens already covered by an earlier selected sentence.

Signal	Weight	How to compute
Weak-character coverage	0.45	Sum the writer's per-character error rate for characters in the sentence.
Confusion-pair coverage	0.35	Reward sentences containing both sides of frequent pairs such as m/n, u/v, 0/O, 1/l.
Medical-token coverage	0.20	Reward uncovered dosage forms, units, frequencies, routes, and number formats.

NAMING

Do not present 'R22-CM' as a published model unless the team can provide the exact paper or repository that defines it. A defensible name for this component is Writer-Specific Confusion-Matrix Calibration (WSCMC).

Evaluation, product flow, and presentation evidence

The demonstration is successful only if the same sealed test crops are evaluated under all model conditions.

1. Required comparison

Condition	Training available	Purpose
A. Generic	No samples from the test writer	Shows the actual starting problem
B. Adapted	E01-E16 only	Measures fixed calibration benefit
C. Adapted + active	E01-E16 and four selected R lines	Measures whether targeted refinement adds value

2. Metrics to report

- OCR: CER and WER, per writer and macro-averaged.
- Structured fields: exact match for medicine, strength, dosage form, frequency, route, and duration.
- Formulary resolution: top-1 and top-3 medicine match accuracy, split into medicines seen and unseen in enrollment.
- Safety: coverage at the chosen confidence threshold, abstention rate, and the number of unsafe false confirmations.
- Efficiency: calibration minutes, adapter training time, adapter size, and inference latency on the target laptop/server.

Product workflow and presentation evidence

3. End-to-end product flow

Clinic staff photographs the prescription. The system corrects perspective, crops lines, recognises text with the clinician's adapter, extracts structured fields, and proposes top-three matches from the clinic formulary. A human confirms or corrects uncertain fields. Only then does the system decrement inventory and create the patient's medication entry. The patient view shows the confirmed medicine, schedule, instructions, history, and reminders. Every correction becomes labelled feedback for a later, auditable model update.

4. What to show in the planning presentation

- The filled calibration sheets and the manifest for at least three pseudonymous writers.
- A side-by-side table of generic, adapted, and actively refined outputs on sealed lines.
- One confusion-matrix visual per writer and an explanation of why the four refinement prompts were selected.
- A working fixed-form capture and crop interface, even if the full clinic dashboard is not ready.
- Architecture for per-writer adapters, human confirmation, formulary matching, inventory updates, patient records, privacy, and audit logs.

5. Starting references

Writer adaptation: MetaHTR (CVPR 2021) and writer adaptation with 16 examples.
Models: Microsoft TrOCR, TrOCR-small, PP-OCRv5, and PaddleOCR custom training.
Indian references: NLEM 2022, PMBJP Product and MRP list, and NFI 2021.
Datasets: IAM, Handwritten Medical Term Corpus, and RxHandBD.

GO / NO-GO RULE

Proceed to the full Pharmacon product if writer adaptation produces a consistent held-out improvement for most writers and the system can abstain on uncertain medicine or dosage fields. If it improves only memorised medicines or only the enrollment sheet, redesign the adaptation before building the dashboard.